YEAR 7 · SCIENCE · MODELS · CHALLENGE HOMEWORK

Why Is Mr Bowen Still Awake?

Half-life with coffee, an interview at home, carbon dating and the Sun



Name: _____ **Class:** _____

Date given: _____ **Date due:** _____

How to do this homework

- **Part A:** Mr Bowen's coffee. It shows you the method.
- **Part B:** interview an adult at home. **Do it at the weekend.**
- **Part C:** carbon dating. **Part D:** the Sun.
- You need a **calculator**, a **ruler**, and a **phone or computer**.
- Answer in **full sentences**. "50 mg" is not a sentence. "He still has 50 mg of caffeine" is.

Remember from the lesson

Half-life: the time it takes for half of something to disappear.

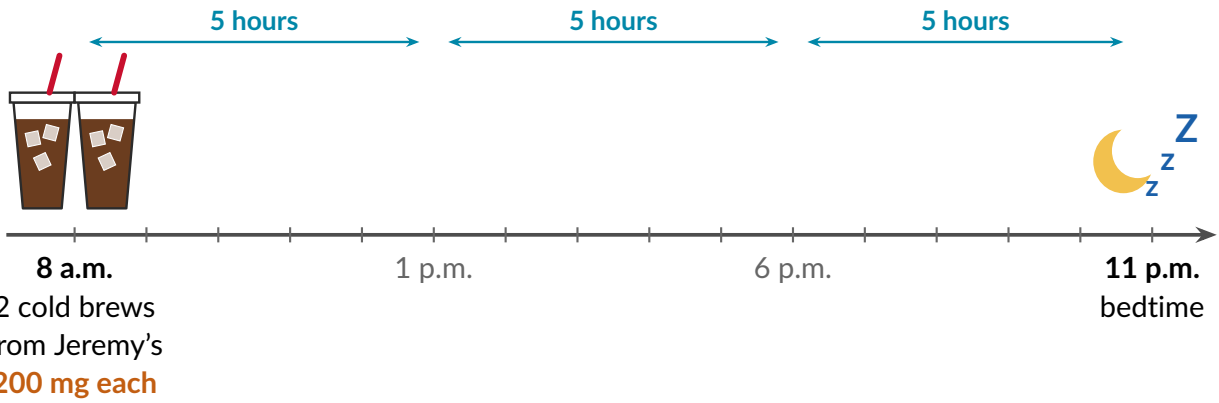
Carbon-14: a rare kind of carbon that slowly breaks down into nitrogen.

Carbon dating: finding the age of something that was once alive by measuring how much carbon-14 is left.

Caffeine's half-life: about **5 hours** Carbon-14's half-life: **5 700 years**

Part A Mr Bowen's Coffee

On Monday, Mr Bowen drinks two cold brews at 8 a.m. He goes to bed at 11 p.m.

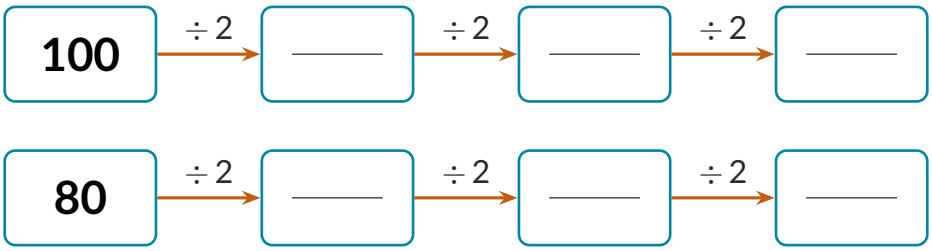


Word help

caffeine	a chemical in coffee, tea, cola and energy drinks. It keeps you awake.
mg	milligram. A grain of rice is about 20 mg.
cold brew	strong coffee made slowly with cold water
halve	cut in half: divide by 2
curve	a line that bends smoothly

A1 Halving

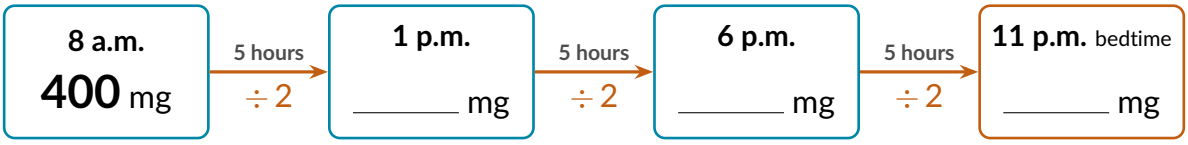
Halve each number.



A2 Mr Bowen's caffeine

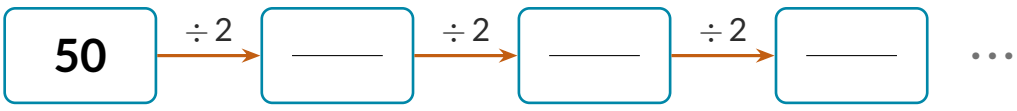
(a) Two cold brews: $2 \times 200 =$ _____ mg

(b) Fill in the boxes. Every 5 hours, **halve** the caffeine.



A3 Does it ever run out?

(a) Keep halving after bedtime.



(b) If you keep halving, does the number ever get to 0? ☐ yes ☐ no

(c) Fill in the gap with one word from the box.

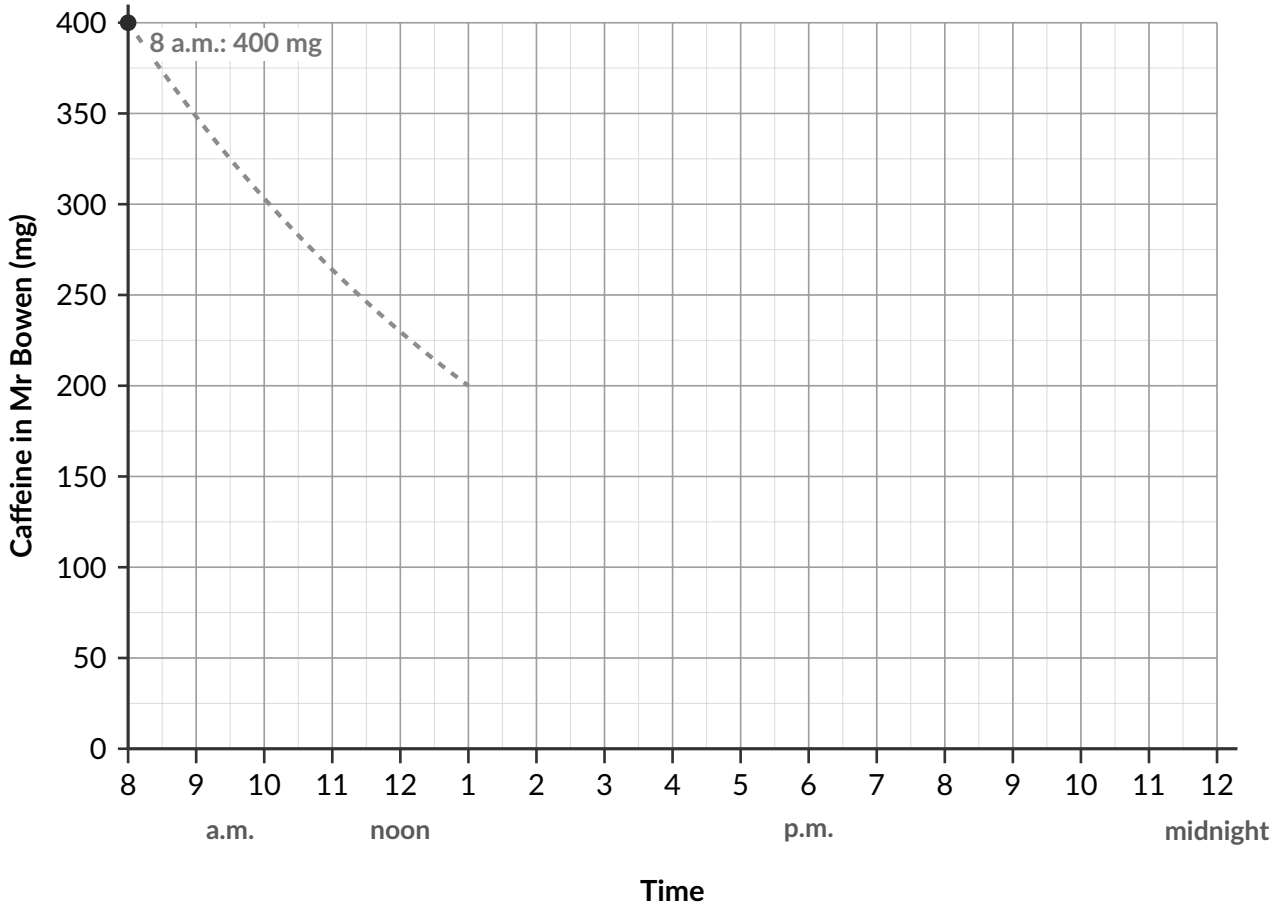
half-life	limitation	molecule

The maths model says the caffeine **never** runs out. But real caffeine is made of molecules, and the last one is gone after about **2 weeks**. So the model is wrong here.

This is a _____ of the model.

A4 Draw the graph

Put a dot for every box in A2. The first part is drawn for you. Join the dots with a **smooth curve**, not straight lines.



A5 Read the graph

Mr Bowen marks your homework at **8 p.m.**

(a) 8 p.m. is between 6 p.m. and 11 p.m. So his caffeine is between _____ mg and _____ mg.

(b) Go **up** from 8 p.m. to your curve, then **across**. How much caffeine is in him at 8 p.m.? about _____ mg

A6 The title question

Why is Mr Bowen still awake at 11 p.m.? Use your number from A2. A cup of coffee has about 100 mg. How much coffee is that?

Mr Bowen is still awake because at 11 p.m. he still has …

Part B Interview a Caffeine Drinker

What you will do

Interview an adult in your family who drinks coffee, tea, cola or energy drinks like Sting. Then work out how much caffeine is in their body at **bedtime**.

You can do the interview in **any language**. Write in **English**.

Word help

interview	ask someone questions, and write down the answers
label	the words printed on a can or bottle. They say what is inside.
search	type words into Google to find something
estimate	a careful guess, using numbers

B1 The interview

Ask about **yesterday**.

1. "Can I interview you for my homework?"
2. "What was your **first** drink with caffeine yesterday? How big was it?"
   For example: 1 can of Sting, 330 ml.
3. "What time did you drink it?"
4. "What time did you go to bed?"

I interviewed my _____.

Their first drink was _____

They drank it at _____.

They went to bed at _____.

B2 Find the caffeine

Where to look

1. **The label** on the can or bottle. Look for the word **caffeine**.
2. **Google.** Search for the drink, like this:
   caffeine in 1 can of Sting mg

Milk, sugar and ice: **0 mg**.

Two numbers, like 50 to 80 mg? Use the **middle**: 65 mg.



Some labels give the caffeine in 100 ml. This bottle has 20 mg in every 100 ml. It holds 300 ml.
300 ml is 3 lots of 100 ml. $3 \times 20 =$ **60 mg** in the whole bottle.

A 330 ml can? $330 \div 100 = 3.3$, then $3.3 \times 20 = 66$ mg. Or search on Google for the whole can.

Their drink has about _____ mg of caffeine.

I found it on: ☐ the label ☐ Google ☐ another website

B3 Halve it, every 5 hours

Start with their drink. In each box, **add 5 hours** to the time and **halve** the caffeine. Keep going until you are **past their bedtime**.

Example: a cup of tea, about 40 mg, at 7 a.m.

	The drink	5 hours later	5 hours later	5 hours later	5 hours later
Time	7 a.m.	12 noon	5 p.m.	10 p.m.	3 a.m.
Caffeine (mg)	40	20	10	5	2.5

Now yours:

	The drink	5 hours later	5 hours later	5 hours later	5 hours later
Time					
Caffeine (mg)					

B4 Bedtime

(a) Their bedtime is between the times in two of your boxes: _____ and _____.

(b) So at bedtime, they have between _____ mg and _____ mg.

(c) Choose a number in between. At bedtime, they have about _____ mg of caffeine.

B5 Compare with Mr Bowen

Mr Bowen had 50 mg at bedtime. Did your adult have **more** or **less**? Why? Think about how **big** the drink was, and **what time** they drank it.

At bedtime my _____ had about _____ mg of caffeine. That is more / less than Mr Bowen, because …

B6 Is your number right?

Your bedtime number is an **estimate**. Tick **one** reason it might be wrong. Then write it as a sentence.

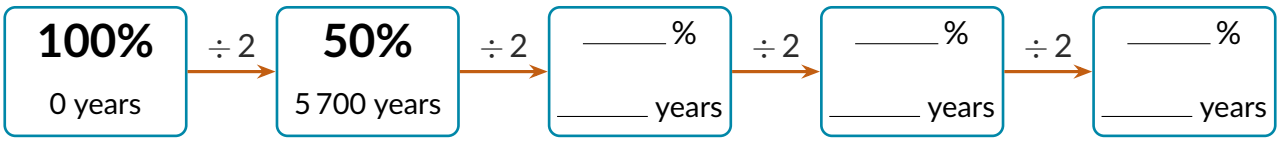
- ☐ They had other drinks with caffeine too.
- ☐ Caffeine's half-life is not 5 hours for everybody.
- ☐ The number from the label or Google may not be exact.

My bedtime number might be wrong because …

Part C Carbon Dating

C1 The halving chain

Caffeine halves every 5 hours. Carbon-14 halves every **5 700 years**. Fill in the boxes.



Each box is 5 700 years more than the one before it.

C2 The museum

Scientists measured the carbon-14 left in five objects. Use C1.

Careful: carbon dating only works for things that were **alive**, up to about **50 000 years** old. If it does not work, write **does not work**.

Object	Was it ever alive?	Carbon-14 left	How many half-lives?	Age (years)
A wooden boat	yes	50%	1	5 700
A bear bone from a cave		25%		
A wooden spear		12.5%		
A stone axe		—		
A *T. rex* tooth		0%		

Word help: a **spear** is a long stick with a sharp point · an **axe** is a tool for cutting wood · a ***T. rex*** is a dinosaur

C3 Why does it not work?

(a) Carbon dating does not work for the stone axe, because stone was never _____ .

(b) Carbon dating does not work for the *T. rex* tooth, because it is too _____ .

Clue: dinosaurs died out about 66 million years ago.

Part D The Sun: Different Speeds, the Same Maths

Remember from the lesson

Caffeine, **carbon-14** and the **Sun's hydrogen** all disappear at a rate we can model with maths. Measure what is **left** to find **how long** it has been.

Word help: the **core** is the middle of the Sun · **hydrogen** and **helium** are gases · a **billion** is a thousand million

D1 How old is the Sun?

Nobody can take a piece of the Sun. So scientists build a **model Sun** with maths, and let it get older until it matches the real Sun.

(a) The Sun turns _____ into _____ .

(b) When the Sun was born, its core was **71%** hydrogen. Half of 71% is _____ %.

(c) Today the core is **34%** hydrogen. Is about **half** of the hydrogen gone? ☐ yes ☐ no

(d) The model Sun matches the real Sun at about _____ billion years. So the Sun is about _____ billion years old.

D2 Three clocks

Draw a line from each clock to how fast it goes.

 **Caffeine**	•	• half gone every 5 700 years
 **Carbon-14**	•	• about half gone in 4.6 billion years
 **Hydrogen in the Sun's core**	•	• half gone every 5 hours

Put them in order, **fastest** first.

1. _____ 2. _____ 3. _____

D3 The same maths

What is the **same** about caffeine, carbon-14 and the Sun's hydrogen? What is **different**?

They are the same because we measure what is left to find … They are different because …

D4 One more link

Caffeine is made of carbon atoms. Where was the carbon in Mr Bowen's coffee made? Tick **one**.

- ☐ In a coffee shop.
- ☐ Inside a star, long before the Sun was born.
- ☐ In Mr Bowen's kitchen.

Before you hand it in: tick each one

- ☐ Every box in A2, B3 and C1 has a number in it.
- ☐ My graph in A4 is a **smooth curve**.
- ☐ B2 says **where** I found the caffeine.
- ☐ A6, B5, B6 and D3 are **full sentences**.